

UNIT-1

MCQs – Data Warehousing (DW) & Business Intelligence (BI)

1. Which of the following best describes the main purpose of Business Intelligence (BI)?

- A) To store large volumes of data
- B) To transform raw data into meaningful insights for decision-making
- C) To perform transaction processing
- D) To replace the need for data warehousing

Answer: B

2. What is the primary function of a Data Warehouse?

- A) Real-time transaction processing
- B) Long-term storage of operational data
- C) Storing integrated, subject-oriented, time-variant, and non-volatile data for analysis
- D) Replacing operational databases

Answer: C

3. Which of the following is NOT a defining feature of a data warehouse?

- A) Subject-oriented
- B) Time-variant
- C) Volatile
- D) Non-volatile

Answer: C

4. What does OLAP stand for?

- A) Online Analytical Processing
- B) Online Application Processing
- C) Offline Analytical Processing
- D) Onsite Analytical Performance

Answer: A

5. Which of the following best describes a Data Mart?

- A) A subset of a data warehouse focused on a specific business area
- B) A replacement for a data warehouse
- C) A transactional database
- D) An operational data store

Answer: A

6. Which of the following best describes Metadata in the context of data warehousing?

- A) Data stored in the warehouse
- B) Data about the data stored in the warehouse
- C) Raw operational data
- D) Real-time transactional records

Answer: B

7. Which of the following is an example of an OLTP system?

- A) Banking transaction system
- B) Sales trend analysis dashboard
- C) Monthly business report generator
- D) Data warehouse query tool

Answer: A

8. Which of the following is NOT part of the BI lifecycle?

- A) Data collection
- B) Data integration
- C) Transaction logging
- D) Data analysis

Answer: C

9. Which of the following best describes Dimensional Analysis in BI?

- A) Analysis of transactions in a normalized database
- B) Viewing data in multiple perspectives such as time, geography, and product
- C) Only storing data in flat files
- D) Maintaining indexes on all columns

Answer: B

10. In the context of BI, “Raw Data to Valuable Information”

means: A) Storing raw operational data directly in a warehouse

B) Processing, cleansing, and summarizing raw data for decision-making

C) Using raw logs without transformation

D) Ignoring redundant data

Answer: B

11. Which of the following is true about OLTP systems?

- A) Optimized for read-heavy workloads
- B) Contain historical data for analysis
- C) Optimized for large numbers of small transactions
- D) Designed for complex analytical queries

Answer: C

12. Which component of a data warehouse is responsible for extracting, transforming, and loading data?

- A) OLAP cube
- B) ETL process
- C) Data mart
- D) Metadata repository

Answer: B

13. Which is an advantage of using Data Warehousing for BI?

- A) Supports operational transaction updates
- B) Provides integrated and consistent data for analysis
- C) Eliminates the need for any data governance
- D) Requires no data transformation

Answer: B

14. Which of the following describes the relationship between BI and DW?

- A) BI replaces DW completely
- B) DW stores the data, BI analyzes it
- C) BI is a form of DW
- D) DW performs reporting while BI stores data

Answer: B

15. The time-variant characteristic of a data warehouse means:

- A) Data is updated in real-time
- B) Data is stored with historical context

- C) Data is erased after a certain period
- D) Data is encrypted before storage

Answer: B

16. Which of the following is a current trend in data warehousing?

- A) Manual data entry
- B) Cloud-based data warehouses
- C) Only on-premises deployments
- D) Avoiding big data integration

Answer: B

17. The “non-volatile” nature of a data warehouse means:

- A) Data changes frequently
- B) Data is not updated or deleted once entered
- C) Data is volatile until analyzed
- D) Data is stored in RAM

Answer: B

18. OLAP differs from OLTP because:

- A) OLAP supports complex analytical queries, OLTP supports day-to-day transactions
- B) OLAP stores operational data, OLTP stores historical data
- C) OLAP is faster for transactions
- D) OLAP uses normalized databases only

Answer: A

19. Which of the following is part of the basic elements of data warehousing?

- A) ETL process
- B) Metadata
- C) Data marts
- D) All of the above

Answer: D

20. In BI, “Reporting and Analyzing Data” typically involves:

- A) Only storing data
- B) Presenting processed data using dashboards, charts, and summaries
- C) Raw data collection without processing
- D) Performing transactions in real-time

Answer: B

5-Mark Descriptive Questions

Short/Descriptive Questions

2 Marks

1. Define Business Intelligence (BI).
2. What is the role of Metadata in a Data Warehouse?
3. Mention two differences between OLAP and OLTP.

3 Marks

1. Explain the relation between Business Intelligence (BI) and Data Warehousing (DW).
2. List and briefly describe the defining features of a Data Warehouse.
3. Differentiate

between Data Warehouse and Data Mart.

4. What are some common OLAP operations used in data analysis?

4 Marks

1. Describe the lifecycle of data in BI.
2. Explain the basic elements of Data Warehousing with examples.
3. Discuss trends in Data Warehousing.

5 Marks

1. Discuss in detail the building blocks of a Data Warehouse.
2. Explain dimensional analysis with a suitable example.
3. Compare and contrast OLAP and OLTP with examples.
4. Describe the typical multitiered architecture of a data warehouse.
5. Explain the differences between star schema, snowflake schema, and fact constellation schema advantages?

UNIT-2

MCQs (1 Mark Each)

Q1. Which of the following best describes *Data Mining*?

- A) The process of cleaning data
- B) The process of discovering patterns and knowledge from large amounts of data
- C) The process of storing data in a database
- D) The process of deleting irrelevant data

Answer: B

Q2. Which of the following is **NOT** a functionality of Data Mining?

- A) Classification
- B) Clustering
- C) Summarization
- D) Data Encoding

Answer: D

Q3. Which classification of Data Mining systems is based on the type of data handled?

- A) Based on data models
- B) Based on kinds of databases mined
- C) Based on applications
- D) Based on degree of user interaction

Answer: B

Q4. The *KDD Process* stands for:

- A) Knowledge Discovery in Data
- B) Key Data Discovery
- C) Knowledge Definition and Design
- D) Knowledge Data Derivation

Answer: A

Q5. Which of the following is a **Data Mining task primitive**?

- A) Specifying the kind of knowledge to be mined
- B) Data cleaning process

- C) Database backup operation
- D) Normalization of data

Answer: A

Q6. Integration of a Data Mining system with a Data Warehouse

- improves:
- A) Data redundancy
 - B) Performance and scalability
 - C) Manual report writing
 - D) Disk fragmentation

Answer: B

Q7. Which is **NOT** an issue in Data Mining?

- A) Mining methodology and user interaction
- B) Performance issues
- C) Security and social issues
- D) Table formatting in spreadsheets

Answer: D

Q8. Which stage in the KDD process involves removing noise and inconsistencies?

- A) Data Selection
- B) Data Cleaning
- C) Data Transformation
- D) Pattern Evaluation

Answer: B

Q9. In classification of DM systems, systems that work without human intervention are classified as:

- A) Interactive DM systems
- B) Autonomous DM systems
- C) Data-centric DM systems
- D) Pattern-oriented DM systems

Answer: B

Q10. Which Data Mining functionality aims to assign data items to predefined categories?

- A) Clustering
- B) Classification
- C) Association
- D) Summarization

Answer: B

1. The main motivation behind Data Mining is:

- A) To create new databases
- B) To extract useful patterns from large data sets
- C) To replace OLTP
- D) To delete redundant data

Answer: B) To extract useful patterns from large data sets

2. Which of the following is NOT a Data Mining task primitive?

- A) Task relevant data
- B) Kind of knowledge to be mined
- C) Concept hierarchy
- D) Query optimization

Answer: D) Query optimization

3. The process of filling missing values with mean, median, or mode is called:

- A) Data Reduction
- B) Data Integration

- C) Data Cleaning
- D) Data Compression

Answer: C) Data Cleaning

4. Dimensionality reduction is mainly used to:

- A) Increase storage space
- B) Reduce the number of features
- C) Create new attributes
- D) Improve metadata description

Answer: B) Reduce the number of features

5-Mark Descriptive Questions

Short/Descriptive Questions

2 Marks

1. Define Data Mining.
2. What is the KDD process?
3. List two data reduction techniques.

3 Marks

1. Explain the functionalities of Data Mining.
2. Differentiate between data cleaning and data integration.
3. What are data mining task primitives? Give examples.

4 Marks

1. Explain various issues in Data Mining.
2. Discuss the role of discretization and concept hierarchy in DM.
3. Describe dimensionality reduction techniques with examples.
4. List the major tasks involved in data preprocessing. Explain the significance of each?
5. What is Histogram ? Explain with example.
6. What is Sampling ? Explain with example.

5 Marks

1. Explain the integration of a Data Mining system with a database or a Data Warehouse.
2. Discuss in detail the phases of the KDD process.
3. Explain data preprocessing steps in detail with examples.

Unit 3: Concept Description and Association Rule Mining

MCQs

1. The process of summarizing data to a higher level is called:
A) Data Reduction
B) Data Generalization
C) Data Cleaning
D) Data Transformation
Answer: B) Data Generalization
2. Market basket analysis is associated with:
A) Classification
B) Association Rule Mining
C) Clustering
D) Regression
Answer: B) Association Rule Mining
3. Which algorithm is widely used for finding frequent itemsets?
A) Decision Tree
B) K-means
C) Apriori
D) SVM
Answer: C) Apriori
4. In association rules, support indicates:
A) Strength of the rule
B) Frequency of occurrence of itemsets
C) Probability of rule accuracy
D) Correlation between classes
Answer: B) Frequency of occurrence of itemsets
5. Incremental ARM is used for:
A) Updating rules dynamically as new data arrives
B) Deleting old itemsets
C) Reducing rule confidence
D) None of the above
Answer: A) Updating rules dynamically as new data arrives

Short/Descriptive Questions

2 Marks

1. What is attribute relevance?
2. Define support and confidence in Association Rule Mining.
3. Mention one difference between classification and association.

3 Marks

1. Explain Market Basket Analysis.
2. What is the role of concept description in Data Mining?
3. Differentiate between Apriori and Improved Apriori algorithms.

4 Marks

1. Explain the process of finding frequent itemsets using Apriori algorithm.
2. Describe associative classification with an example.
3. Discuss class comparisons in concept description.

5 Marks

1. Explain in detail the Apriori algorithm with an example.
2. Discuss incremental association rule mining and its advantages.
3. Write a note on rule generation and evaluation in ARM.